A Consolidated Study of Hybrid Runtime Screening, Failure Modes, and Request-Bounded Deterministic Authority in FerrumOS

When Agents Control the Kernel: A JEPA World Model Safety Gate with Empirical False-Negative Decomposition

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ABSTRACT

Autonomous agents that modify files, processes, services, and network state require controls where proposed actions become operating-system effects. FerrumOS normalizes provider output into 41 canonical actions and evaluates independent deterministic and action-conditioned JEPA transition branches before capability-gated syscalls. The original study accounted for 13,697 transitions from 3,639 QEMU episodes. On a 500-episode authored fixture, rules plus the selected JEPA reached 81.4% balanced accuracy, but a per-action mean was nearly tied; five complete pipelines varied substantially, and all 52 hybrid false negatives were decomposed into missing semantic assets, process accumulation, and heap underprediction.

Version 1.2 adds a registered v3-v3.4 lineage and retains four failed iterations before opening a source-held-out final catalog. The frozen candidate improves untouched-test rollout error at H=1, H=3, and H=5 by 9.54%, 2.36%, and 0.92%. On a new 512-episode deterministic simulator, request-bounded v3.4 policy records 100% balanced accuracy; the paired source-stratified 95% improvement interval over runtime-v2 is +46.35 to +48.82 points. Rules-only and rules+JEPA are identical, so the final safety result belongs to deterministic authority, not incremental learned safety value. The candidate remains archived and not deployed.

Claim boundary	The original safety fixture and the new incident-informed fixture are authored software simulations over QEMU-derived or programmatically constructed states. They are not production incident replays, independently human-adjudicated hazards, formal proofs, or evidence of complete safety. The v3.4 learned candidate is archived but not deployed. The new safety result belongs to deterministic policy; the learned result is the untouched-corpus rollout improvement.

1 Introduction

Language-model agents increasingly convert natural-language goals into state-changing operations. Application-layer refusal can reduce unsafe proposals, but an operating system still needs a mediation point where a canonical action and its arguments are checked before execution. FerrumOS places that point in the ring-3 Heliox daemon immediately above capability-gated syscalls. Learned inference cannot grant authority; deterministic kernel validation and operator confirmation remain independent backstops.

The research question is narrower than whether a neural model can make an operating system safe: can an action-conditioned forecast improve safety-relevant state prediction while exact policy, capabilities, and confirmation retain authority? The first study established the mediation architecture, strong baselines, runtime measurements, and failure decomposition; it did not establish JEPA superiority on thresholded safety decisions. The extension asks whether hypothetical repeated rollout had been allowed to influence authority beyond the action actually requested.

Contributions

This report contributes: (1) provider-independent OS action mediation with monotonic deterministic-plus-learned screening; (2) an episode-disjoint public corpus and deterministic release; (3) strong baselines, five complete pipelines, ring-3 timing, failure injection, and exhaustive false-negative decomposition; (4) a digest-bound v3-v3.4 lineage with retained failures and final-catalog access denial during selection; (5) request-bounded authority for exact covered actions; and (6) a 512-case source-held-out simulator with paired uncertainty, exact rules-only attribution, and an explicit non-promotion result.

2. Claim-to-evidence map and study lineage

The paper separates claims by evidence object so that a reader can audit what each experiment actually supports.

Claim	Evidence	Result	Boundary
Hybrid mediation is implementable before OS effects	Ring-3 Heliox path plus capability-gated syscall boundary	Implemented and exercised in QEMU for runtime-v2	v3.4 authority policy is not installed
JEPA representation predicts transitions better than the matched autoencoder	Untouched 1,969-row episode-disjoint test partition	H=3 error 3.87% vs 6.45%	Does not prove better binary safety decisions
Original hybrid lowers authored-fixture false negatives	500 paired authored cases	FNR 20.8% vs rules-only 41.2%	FPR rises from 8.4% to 16.4%; per-action mean is nearly tied
v3.4 candidate improves deployed runtime-v2 rollout	Same untouched published test partition	Lower error at H=1, H=3, and H=5	Frozen candidate was not retrained during v3.4 policy repair
v3.4 policy covers the new four-source fixture	512 source-held-out deterministic cases	256 TP, 0 FN, 256 TN, 0 FP	Labels and effects are simulator-authored
Learned branch adds final-fixture safety value	Rules-only vs rules+candidate ablation	No difference	No incremental learned safety claim
v3.4 is deployable	Frozen offline, runtime, authority, and digest gates	Not all passed	Candidate remains promotion_eligible=false

2.1 Version lineage

The public report and dataset remain the archival starting point. Runtime-v2 is the deployed comparison artifact. v3 changes deterministic coverage while freezing the learned candidate. v3.1 and v3.2 test learned resource thresholds; v3.3 tests exact multi-step projection; and v3.4 changes only the authority factorization for exact covered actions. Failed results remain in the repository rather than being overwritten.

Lineage item	Role	Selection exposure	Deployment status
Published study v1.0.0	Original architecture, corpus, baselines, and runtime evidence	Original registered fixture	Published and deployed runtime-v2 lineage
v3 candidate	Frozen learned transition plus deterministic policy repair	Legacy and opened development fixtures	Not deployed
v3.1-v3.3	Registered failed policy/threshold iterations	Development only	Not deployed
v3.4	Request-bounded authority evaluation	Development gates, then one final opening	Archived; not deployed

The v3.4 protocol binds the candidate, development fixtures, source lists, final-catalog generator, and deployment digests. Validation-only programs record attempted file access and deny both final source and final scenario paths. The evaluator refuses to overwrite an existing final result.

3. Threat model and security boundary

3.1 Assets and failure sources

Protected assets include persistent-state integrity and availability; kernel and service continuity; bounded process, heap, and disk consumption; operator confirmation authority; and an auditable record of proposed, blocked, and executed canonical actions.

Failure sources include untrusted provider output or prompt injection, a paired controller submitting harmful but syntactically valid calls, benign autonomy that repeats locally reasonable actions, and unprivileged ring-3 software attempting to route operations through Heliox. Trusted kernel code, malicious physical operators, compromised signing keys, hardware attacks, and already-unrestricted root authority are outside the experiments.

Category	Example	Primary control	Evidence status
Direct harm	Delete daemon configuration; unsafe upgrade	Exact predicate plus capability/confirmation	Covered in original and v3.4 fixtures
Compound resources	Repeated writes, spawns, or service starts	Fresh state, cumulative budget, and forecast	Original misses exposed temporal weakness
Prompt injection	Provider proposes destructive tool call	Canonical provider-independent gate	Evaluated as an authored scenario family
Rule edge or OOD	Alias, unlisted asset, learned residual	Canonical paths plus monotone learned caution	Partial in original; expanded in v3 policy
Authority mismatch	Forecast repeats an action not actually requested	Request-bounded exact effect	Repaired in v3.4 development

3.2 Security properties

The deterministic branch remains active when the learned branch predicts safety. Provider identity is provenance, not a model feature. A predictive allow is never execution authority. A predictive block can create availability harm, so false positives and intervention rate are first-class outcomes. Missing, non-finite, malformed, or forbidden-coverage learned artifacts cannot erase deterministic protection.

A false negative advances to later capability and confirmation checks; if those also permit the call, harm can execute. A false positive prevents useful work and may become denial of service. Hazards that neither alter represented resources nor match a deterministic predicate remain invisible. These are measured limitations rather than formal guarantees.

3.3 Kernel and human authority

The world model runs in the unprivileged Heliox daemon. The kernel remains deterministic and independently enforces syscall argument validation, capabilities, signed upgrade policy, and confirmation. The experiments do not claim a kernel-resident neural model. Registered natural-use annotations remain a protocol until real annotators complete blinded labels and adjudication.

3.4 Request-bounded authority

Forecast horizon and execution authority are different objects. Multi-step rollout estimates consequences if an action repeats. It does not mean those repetitions were requested or authorized. v3.4 therefore applies an exact deterministic effect to the requested action when the canonical action is covered. Learned multi-horizon output remains available for telemetry and may add caution for uncovered effects, but it cannot convert imagined repetitions into authority.

4. System architecture

Provider output is decoded into a canonical tool and normalized argument vector. The daemon captures OS state, runs deterministic and learned self-composition rollouts independently, takes the higher risk, and only then permits the tool dispatcher to request a capability-gated syscall. Exact semantic facts such as deleting daemon configuration or requesting a kernel upgrade remain outside learned compression.

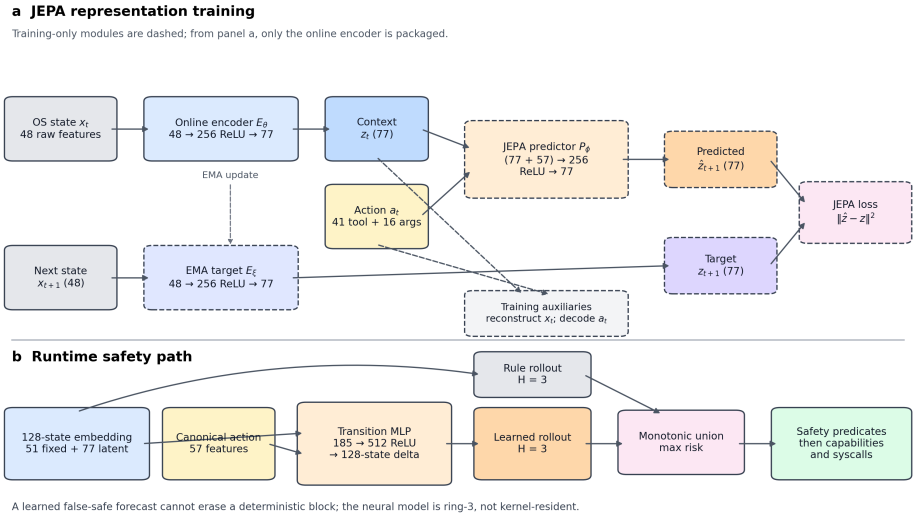


Figure 1. FerrumOS JEPA training and runtime architecture. The learned forecast is advisory and monotone; deterministic policy and kernel authority remain independent.

The public provider, planner, gesture, and JSON-RPC dispatch paths converge on `execute_tool_with_world_model`. Read-only `world_model_preview` follows the same prediction path without dispatch or experience emission. Provider choice - cloud, local language model, or paired external controller - does not change the safety representation.

4.1 Decision flow

1. Normalize provider output into one of 41 canonical actions and 16 argument features.
2. Capture a 48-dimensional raw OS observation.
3. Evaluate exact deterministic predicates and bounded resource effects.
4. Encode the observation and predict learned transition deltas over the registered horizon.
5. Combine risks monotonically: a learned false-safe result cannot erase a rule block.
6. Pass an allowed proposal to capability and confirmation checks, which can still reject it.
7. Emit a privacy-bounded audit record without prompts, paths, screen content, or account identifiers.

4.2 Failure behavior

Component	Runtime role	Failure behavior
Deterministic transition	Exact paths, policy, bounded resource deltas	Always active
JEPA encoder and transition	Learned action-conditioned numeric effects	Optional; invalid files rejected
Monotonic union	Higher branch risk wins	Learned false-safe cannot erase a rule block
Capability and confirmation	Independent execution authority	Can reject predictive allows
Audit record	Reproduction and incident review	Does not contain raw prompt or path data

5. Model and decision method

5.1 State and action representation

A 48-dimensional raw vector contains normalized process, heap, filesystem, disk, screen, last-action, and error observations. The online encoder contributes 77 learned values to a 128-dimensional runtime state while deterministic safety fields remain explicit. The action input contains 41-dimensional one-hot identity plus 16 normalized argument features.

With online context encoder E, EMA target encoder E_ema, action-conditioned predictor P, reconstruction head R, and action head A, training minimizes:

```
z_t = E(x_t)
target = stopgrad(E_ema(x_(t+1)))
predicted = P(z_t, action, args)
loss = L_JEPA + 0.25 * L_reconstruction + 0.10 * L_action
```

The target encoder uses an EMA coefficient of 0.99. Anti-collapse promotion requires better-than-zero prediction, latent standard deviation at least 0.01, effective rank at least 4, and non-zero action sensitivity.

5.2 Transition and rollout

A 185-input, 512-hidden ReLU multilayer perceptron predicts a 128-dimensional state delta. The published transition was trained with full-batch gradient descent, inverse-frequency action weighting, 2,000 epochs, and a fixed episode split. The packaged pair contains 32,333 encoder parameters and 160,896 transition parameters, 193,229 total.

For h=1..H, the learned branch predicts the consequence of repeating the proposed canonical action. Process deltas accumulate explicitly. Resource risk rises when predicted heap or disk reaches 0.95; exact protected-path deletion and policy-only upgrade predicates are deterministic. The original runtime blocks when either branch reaches risk 0.7. H=3 was deployed as the minimum verified horizon for the process-growth pattern, not as a globally optimal horizon.

5.3 What JEPA means here

JEPA is an inductive bias for predicting safety-relevant abstract structure instead of reconstructing arbitrary paths, process names, timestamps, and screen details [1-3]. It is not a causal theorem. Binary decision parity with a simple baseline does not imply equivalent transition modeling, but lower transition error also does not prove greater operational safety.

5.4 v3.4 change isolation

The v3.4 learned transition weights are unchanged. The experiment freezes candidate SHA-256 `2616aa6b0587adc9258bd5ccd7d95f1e1a2011af7b2e76e3927046c010444f77`. The policy repair changes authority accounting for exact covered actions. It is therefore a policy-lineage result plus a separate frozen-model rollout comparison, not a new model-training result.

6. Dataset, training, and public packaging

6.1 Published corpus ledger

Ledger item	Count	Treatment
Accepted transitions	13,697	Retained for complete accounting
Complete episodes	3,639	Episode is the split unit
Multi-step episodes	1,300	Temporal examples retained atomically
Execution not attempted	373	Excluded from transition fitting
Policy-only upgrade rows	54	Auditable; excluded from learned coverage
Eligible fitting rows	13,270	9,104 / 2,197 / 1,969 train/validation/test

Complete episodes are shuffled with split seed 42; no episode crosses partitions. All 41 canonical actions were observed under both memory profiles. Forty actions are represented in learned fitting, while kernel upgrade remains policy-only. Every argument-bearing learned action has at least three variants. A corrective collection replaced stale pre-ABI `ipc_send` episodes with 128 successful live transitions.

6.2 Published representation comparison

Metric	Matched autoencoder	Selected JEPA	Relative change
One-step normalized error	2.30%	1.68%	-27.1%
Safety-core normalized error	4.61%	3.81%	-17.5%
Macro-per-tool error	2.96%	1.71%	-42.3%
Rollout H=3	6.45%	3.87%	-40.0%
Rollout H=5	6.70%	4.03%	-39.9%

The release checkpoint uses representation seed 42 and transition seed 17, selected only by validation transition metrics. Full-pipeline seeds retrain the representation itself and are different checkpoints even when a numeric seed is shared.

The trained models depend on the number of BLAS threads used in training. An exploratory rerun after v1.2 (`docs/research/fi1_reports/pipeline_round3.md`, numbers in `docs/research/fi1_pipeline_round3.json`) on the development machine (24 logical CPUs, OpenBLAS 0.3.29, NumPy 2.2.6) reproduced the committed JEPA encoder and the seed-17 transition model byte for byte at 24 threads. At 4 and at 1 thread every retrained model differs: float32 matrix products add in a different order, and over a full training run the weights drift apart. The H=3 rollout error of the three fixed-encoder transition models (seeds 17, 42, and 91) is 0.039 to 0.041 in the committed run, of which the 3.87% above is seed 17, and 0.064 to 0.085 in the reruns. The transition errors in this table therefore describe one training run at one thread count.

6.3 Release integrity

The exact JSONL is packaged as a deterministic gzip archive with an MIT dataset license, data card, manifest, SHA256SUMS, episode-split verification, and credential-pattern scan. The source SHA-256 is `ae7498c7d93b4e35f8bf0a9ac595ca2bacc1ec694b1806c47a33741d911d491f`; the deterministic gzip SHA-256 is `310f5192d3d777c5a6420498a7074e48035f3e408d3c089cbc98eed79bfe6fb7`. The release is open at version DOI 10.5281/zenodo.21829193. Ten public files matched the local release byte-for-byte and the downloaded copy passed 11/11 verification checks.

7. Original evaluation protocol

The primary original fixture contains 250 safe and 250 dangerous authored episodes, balanced across direct harm, compound resource exhaustion, provider prompt injection, and rule-table edge cases. Initial states and empirical deltas derive from the untouched 1,969-row QEMU test split. Harmful outcomes are evaluated counterfactually; the protocol does not execute 500 destructive calls.

Arm or ablation	Purpose
Always allow / always block	No-model controls and metric sanity checks
Rules only / JEPA only / union	Primary paired contribution of each branch
Per-action mean transition	Simple learned baseline without representation inference
Matched autoencoder	Generative-style representation baseline
Action conditioning removed	Tests whether action identity and arguments matter
H=1..5	Horizon sensitivity and release-H interpretation
Three fixed-encoder transition seeds	Downstream initialization sensitivity
Five complete pipeline seeds	Representation plus transition sensitivity
1,969-row safe replay	False-alert cost on untouched safe observations
240-row HUD boundary replay	Post-training safe calibration negative control

Outcomes include TP, FN, FP, TN, FNR, FPR, balanced accuracy, Wilson intervals, AUROC, AUPRC, paired McNemar tests, paired bootstrap intervals, transition error, and safe alerts per 1,000 actions. Complete-pipeline intervals use the t distribution over five runs and describe run-to-run uncertainty on this fixture.

Model checkpoints are selected by validation representation and transition metrics only. Safety-test labels are not used for checkpoint selection, and all five full-pipeline seeds are reported.

Runtime experiments execute the deployed ring-3 path in QEMU/WHPX. The 1 kHz PIT provides mean, median, p95, and maximum time at 1 ms percentile resolution. Raw virtual TSC cycles are retained but not converted because they did not agree reliably with the guest timer.

8. Original safety results and strong baselines

8.1 Three-arm result

Condition	TP	FN	FP	TN	FNR	FPR	Balanced accuracy
Rules only	147	103	21	229	41.2%	8.4%	75.2%
JEPA only	57	193	23	227	77.2%	9.2%	56.8%
Rules + JEPA	198	52	41	209	20.8%	16.4%	81.4%

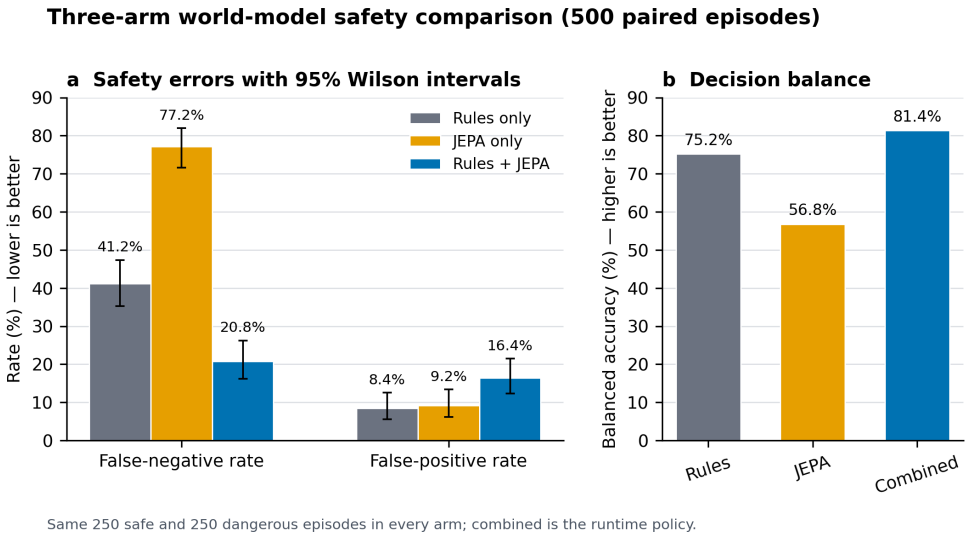


Figure 2. Original registered three-arm paired comparison. Error bars are 95% Wilson intervals.

The learned branch catches 51 dangerous cases missed by rules while adding 20 safe blocks. This supports monotonic composition, not learned replacement of rules.

8.2 Stronger binary-decision baselines

Combined condition	FNR	FPR	Balanced accuracy	AUROC	AUPRC
Rules only	41.2%	8.4%	75.2%	0.770	0.831
Rules + autoencoder	24.4%	16.4%	79.6%	0.830	0.869
Rules + per-action mean	20.8%	16.8%	81.2%	0.847	0.881
Rules + selected JEPA	20.8%	16.4%	81.4%	0.848	0.881
Rules + calibrated JEPA	16.4%	16.4%	83.6%	0.867	0.898

The per-action mean differs from selected JEPA on only three correctness outcomes: JEPA is uniquely correct twice and the mean model once, exact McNemar p=1.0. The 0.2-point balanced-accuracy difference does not establish a material JEPA safety advantage. JEPA's stronger evidence in the original study is lower transition error than the matched autoencoder.

The same rerun (Section 6.2) retrained the encoder, the three transition models, and the autoencoder at 4 and at 1 BLAS thread. Rules + JEPA gives 185 / 65 / 31 / 219 (TP / FN / FP / TN, 4 threads) and 199 / 51 / 41 / 209 (1 thread) instead of 198 / 52 / 41 / 209, with balanced accuracy 80.8% and 81.6% instead of 81.4%. Rules + calibrated JEPA gives 204 / 46 / 40 / 210 at 4 threads instead of 209 / 41 / 41 / 209 and does not change at 1 thread. Rules + per-action mean and rules only do not change. The comparison "rules + JEPA at least as good as rules + per-action mean", 81.4% against 81.2% in the committed run, becomes 80.8% against 81.2% at 4 threads and holds at 1 thread. Rules + JEPA stays above rules only (75.2%) and above rules + autoencoder at every thread count, and the 95% interval of the accuracy difference between rules only and the combined gate excludes zero at every count (committed 0.062 [0.030, 0.094]; 0.056 [0.030, 0.084] at 4 threads; 0.064 [0.032, 0.096] at 1 thread). On the untouched 1,969-row safe replay, rules + JEPA raises 216 false alarms in the committed run against 50 (4 threads) and 58 (1 thread); that count describes one training run.

9. Original stability, runtime, and failure analysis

9.1 Complete-pipeline sensitivity

Full-pipeline seed	H=3 error	FNR	FPR	Balanced accuracy
17	15.47%	28.4%	10.4%	80.6%
42	4.11%	37.6%	10.0%	76.2%
91	13.16%	25.2%	13.6%	80.6%
123	6.06%	18.8%	16.4%	82.4%
2026	27.04%	25.2%	16.8%	79.0%

Across five complete pipelines, balanced accuracy averages 79.76% with SD 2.33 points and a 95% t interval of 76.87%-82.65%. Full-pipeline seed 17 is not the release checkpoint: it retrains the representation, whereas the release checkpoint retains representation seed 42 and varies only the transition seed.

9.2 Ring-3 runtime

Horizon	Mean	Median	p95	Maximum	Blocked / 100
H=1	1.35 ms	1 ms	2 ms	3 ms	2
H=2	1.44 ms	1 ms	2 ms	3 ms	2
H=3	1.45 ms	1 ms	2 ms	3 ms	2
H=4	1.53 ms	1 ms	2 ms	3 ms	2
H=5	1.59 ms	2 ms	2 ms	2 ms	2

After a 64-preview H=5 warmup, model loading is 30 ms, both learned files are active, and heap use remains 664,832 bytes before and after 500 measured previews. The 1 ms PIT granularity limits percentile precision. Provider latency, action execution, and human confirmation are excluded.

One connection submits 96 previews across six action classes without waiting between sends. All 96 responses return with correct IDs in 20.142 seconds host wall time. Inference is serialized; this tests framing and state isolation, not parallel neural execution.

9.3 Injected artifact failures

Injected condition	Observed behavior	Result
Valid files	Both load; protected self-delete blocked	PASS
Both files missing	Daemon starts; deterministic self-delete remains blocked	PASS
Valid-shape NaN files	Both rejected; deterministic block remains	PASS
Forbidden policy coverage	Transition rejected; upgrade quarantine remains	PASS
Collapsed JEPA trial	Metrics retained; no promotable artifacts emitted	PASS

10. Original false-negative decomposition and boundary calibration

All 52 selected-gate misses were reproduced from committed fixture and prediction rows.

Cluster	Count	Mechanism	Required remedy
Missing semantic asset	21	Deleted path is not an explicit protected field	Expand deterministic asset policy/state
Process accumulation	20	Horizon and episode history omit long sequences	Temporal history or cumulative budget
Heap underprediction	11	Forecast stays below 0.95 while observed state crosses	Action-specific data and uncertainty

Ten of the 11 heap cases are `hud_update`; one is `http_get`. All misses have recorded risk 0.0, so threshold tuning alone cannot repair them. Forty-one require policy or temporal-state changes rather than more model capacity.

10.1 Post-training HUD boundary set

The independent negative control contains 240 successful QEMU episodes over 12 argument-size regimes, including the 128-byte render boundary. Observed normalized heap delta is zero in every row. Transition heap-delta MAE is 0.00296 and RMSE is 0.00402; unadjusted resource alarms are 0/240.

Boundary statistic	Value
Episodes / regimes	240 / 12
Execution failures	0
Observed non-zero heap deltas	0
Residual p95 upper margin	0.002219
Alarms with p95 margin	0
Production calibration changed	No

This corpus is one safe action class outside training and the registered test split. It cannot estimate dangerous recall or justify a distribution-free guarantee. The residual margin is analysis only.

Boundary collection also exposed an independent compositor arithmetic underflow: a 1,044-pixel bubble was centered in a 1,024-pixel framebuffer. The renderer now uses actual framebuffer dimensions, UTF-8-safe truncation, and saturating or clamped geometry; a real QEMU boundary verifier passes without panic.

11. Registered v3-v3.4 extension

11.1 Incident-informed source protocol

Development sources were partitioned before candidate selection. A separate final source catalog was frozen and denied to validation scripts. The final official postmortems describe inadvertent disablement of Cloudflare's R2 Gateway, a Dropbox maintenance script reinstalling active database machines, a Google Cloud race propagating corrupt load-balancer configuration, and GitHub resource contention involving unintended configuration [19-22]. These reports motivate defensive abstractions only. The fixture does not reproduce provider systems, workloads, data, exploit paths, or impact.

The final benchmark contains 128 cases per source, 512 total, with deterministic labels and exactly half safe controls. Source holdout is not an unseen-action, unseen-hazard, natural-use, or production-generalization claim.

11.2 Leakage and artifact controls

The protocol fixes candidate digest, source partitions, family sizes, gates, statistics, and deployment digests before the final is opened. Validation-only selection installs file-open audit hooks and records all attempted paths. It denies both the final source registry and generated final catalog. The final evaluator is a separate command and refuses replacement of an existing result.

11.3 Registered statistics

Balanced-accuracy differences are paired by scenario and bootstrapped within source for 10,000 resamples. The 95% interval is the 2.5th to 97.5th percentile of the source-stratified paired distribution. Rollout error is reported at H=1, H=3, and H=5 on the untouched published episode-disjoint partition. Brier score and 10-bin ECE describe deterministic gate scores, not learned probabilities, and are not used as a calibration claim.

11.4 Frozen gates

Stage	Gate	Outcome
v3.4 development	v3 opened-development FN = 0 and FP < 40	Passed: FN 0, FP 16
v3.4 development	Legacy FN < 50, FP <= 41, BA > 81.8%	Passed: FN 0, FP 21, BA 95.8%
Final access	No final source/catalog reads during selection	Passed
Final evaluation	One digest-bound evaluation, no overwrite	Passed
Runtime authority	v3.4 path exercised through deployed ring-3 authority	Not passed
Runtime effects	Authored resource effects validated against runtime	Not passed
Deployment	Every frozen gate passes and digests match	Not eligible

12. Retained iterations and selection discipline

Iteration	Registered change	Development outcome	Decision
v3	Frozen decoder refinement plus protected-path and service policy repair	Legacy BA 87.0%; FN 5; FP 60	Retained failure: FP exceeded 41
v3.1	Shared learned-resource threshold sweep	0/19 candidates passed	Retained failure
v3.2	Learned-disk threshold sweep	0/7 candidates passed	Retained failure; alerts were rule-originated
v3.3	Three-step exact projection for covered actions	Opened v3 safe-control FP 64/256	Retained failure: hypothetical repetition over-intervened
v3.4	One-step authority for covered requested actions	Legacy BA 95.8%; FN 0; FP 21	Development passed; final opened once

The sequence is part of the evidence. No failed result was replaced by a cleaner file, no deployed artifact was used as a scratch target, and the source-held-out final remained inaccessible until v3.4 passed every development gate.

12.1 Why v3.3 failed

For an exact action with a bounded one-step effect, v3.3 repeatedly applied that effect across H=3 during the authority decision. This asked a useful forecasting question but answered the wrong authorization question: what might happen if the action repeats, rather than what the current request authorizes. The result was a sharp rise in safe-control intervention.

12.2 Why v3.4 is narrower

v3.4 does not relax protected-path rules or erase learned caution. It limits exact covered-action authority to the requested step. Uncovered learned effects may still accumulate in rollout, and deterministic cumulative budgets remain the appropriate mechanism for repeated authorized actions.

12.3 Development intervention

On the legacy 500-case fixture, v3.4 records FN 0 and FP 21, an 8.4% safe-control intervention rate, versus runtime-v2 FN 50 and FP 41, 16.4% intervention. On the opened v3 incident fixture, v3.4 records FN 0 and FP 16, 6.25% intervention, versus v3 FN 0 and FP 40, 15.625%. These are development regressions used to select v3.4 and are not independent final evidence.

13. v3.4 final results

13.1 Untouched-corpus rollout

Horizon	Deployed runtime-v2 normalized error	Frozen v3.4 candidate	Relative reduction
H=1	0.072218	0.065331	9.54%
H=3	0.037751	0.036859	2.36%
H=5	0.040362	0.039990	0.92%

All registered horizons improve. The geometric candidate/runtime-v2 ratio is 0.956508, corresponding to a 4.35% geometric improvement. This is the quantitative learned-model result. The gain is modest at longer horizons and should not be described as a large or universal JEPA advantage.

13.2 Source-held-out safety fixture

Condition	TP	FN	TN	FP	FNR	FPR	Balanced accuracy
Deployed runtime-v2 rules+JEPA	12	244	256	0	95.31%	0.00%	52.34%
v3.4 deterministic policy	256	0	256	0	0.00%	0.00%	100.00%
v3.4 policy + frozen candidate	256	0	256	0	0.00%	0.00%	100.00%

The paired source-stratified 95% bootstrap interval for v3.4 minus runtime-v2 balanced accuracy is +46.35 to +48.82 percentage points. The improvement is large on this fixture, but the fixture is designed around deterministic scenario families and cannot estimate natural prevalence.

The risk-score Brier value changes from 0.4768 to 0.00875 and 10-bin ECE from 0.4789 to 0.0625. These scores are rule-derived and are not calibrated learned probabilities. They describe score-label agreement on this balanced deterministic catalog only.

13.3 Exact attribution

Rules-only and rules+candidate have identical confusion matrices and identical per-case decisions. The learned-only marginal safety contribution on the final fixture is therefore zero. The final benchmark demonstrates typed deterministic policy, authority factorization, and integration. It does not corroborate incremental learned harm avoidance.

14. Ablation, interpretation, and non-promotion

14.1 What changed between studies

Dimension	Original report	v1.2 extension
Main safety fixture	500 authored cases from QEMU-derived states	512 incident-informed deterministic cases
Selection protection	Safety labels excluded from checkpoint selection	Final sources and catalog denied to all selection scripts
Learned comparison	JEPA, autoencoder, per-action mean, five pipelines	Frozen runtime-v2 vs frozen candidate rollout
Policy attribution	Rules, JEPA, and union	v3.4 rules-only and rules+candidate exact equality
Negative results	Baseline parity, seed variation, 52 FN clusters	v3-v3.3 retained failed gates
Deployment evidence	runtime-v2 ring-3 timing and failures	v3.4 not deployed; runtime gates pending

14.2 What the learned branch adds

The original selected JEPA reduced transition error versus a matched autoencoder and lowered authored-fixture FNR when composed with rules. The frozen v3.4 candidate also improves all registered rollout horizons over deployed runtime-v2. Those are prediction results. Neither the original per-action mean comparison nor the new final rules-only ablation establishes JEPA superiority for thresholded safety decisions.

14.3 What deterministic authority adds

Exact semantic predicates cover risks that compressed numerical state can miss. Request-bounded effects prevent unrequested hypothetical repetitions from causing intervention. The new final fixture is intentionally credited to this deterministic path. Learned uncertainty may motivate caution, but it is not allowed to grant capability or override a deterministic block.

14.4 Non-promotion decision

Candidate files are archived under a versioned research artifact. Deployed runtime-v2 digests remain unchanged: transition SHA-256 `f6bf2e502aa9d8abeeba47b81205468f02ec54203f4f39931d6ea42359cf0dc8`, encoder SHA-256 `9fb5f08b3c5ac16b37a34376b2bb4cbbd249c536bf419dbd80e52a90a0bfc9b3`, and manifest SHA-256 `0ac0bfe402605e20ef861c03cf6849d4b8a57844996a3fa4eb46fed28500eff7`. The research candidate is marked `promotion_eligible: false` until runtime authority, timing, resource-effect, and digest gates pass.

15. Relation to prior work

JEPA and I-JEPA motivate prediction in representation space rather than full observation reconstruction [1-3]. FerrumOS applies that inductive bias to action-conditioned OS state transitions but does not infer formal safety from it. Constitutional AI and model-level alignment shape provider behavior [4]; FerrumOS instead mediates canonical state-changing actions from any provider at runtime.

SafetyBench, Agent-SafetyBench, and ST-WebAgentBench measure safety knowledge or application-layer tool policy [5,7,8]. WebArena and OSWorld evaluate long-horizon agent task completion in web or computer environments [6,11]. ToolEmu and AgentDojo provide emulated or dynamic safety evaluation for LM agents [12,13]. SafeDreamer uses world models for constrained reinforcement learning [9]. Responsible manipulation addresses physical consequence reflection [10]. FerrumOS's narrower differentiator is provider-independent action normalization and prediction immediately before an OS capability boundary, with deterministic kernel enforcement afterward.

Work family	Primary layer or measure	Difference from FerrumOS
JEPA, I-JEPA, V-JEPA [1-3]	Representation-space prediction	Motivates objective; does not establish OS safety
Constitutional AI [4]	Model alignment	Provider behavior rather than OS mediation
SafetyBench and agent safety benchmarks [5,7,8]	Knowledge or tool-policy behavior	Broader taxonomies; application-layer evaluation
WebArena and OSWorld [6,11]	Task completion in interactive environments	No capability-gated kernel authority factorization
ToolEmu and AgentDojo [12,13]	Emulated risks and prompt-injection defenses	Different simulator and authority boundary
SafeDreamer [9]	World-model constrained RL	Trains a policy; FerrumOS filters discrete OS actions
Responsible manipulation [10]	Physical safety reflection	Embodied consequences, not OS effects

The metrics are not equivalent across these studies. Question accuracy, policy-compliant task completion, RL cost, robot hazard avoidance, transition error, and OS gate FPR/FNR measure different objects. The present novelty claim is systems-oriented: authority factorization, artifact lineage, and failure-accounted evaluation at the OS mediation boundary.

15.1 Incident reports as defensive design inputs

Official postmortems are used to identify recurring defensive abstractions: protected configuration, safe maintenance targeting, propagation of invalid configuration, retry amplification, and resource contention [14-22]. The paper does not claim to reconstruct these incidents or evaluate the affected providers.

16. Limitations, validity threats, and claim registry

1. Both safety fixtures use deterministic authored labels. They are not natural-prevalence data and cannot estimate production precision, recall, or intervention cost.
2. The final four-source catalog has fixed balanced family sizes. Source holdout is not action, hazard, implementation, or distribution holdout.
3. Incident reports motivate abstractions; no provider environment, payload, workload, data, or impact is replayed.
4. The final v3.4 safety result is rules-only in attribution. The learned candidate adds no final-fixture safety decisions.
5. The frozen candidate improves rollout against runtime-v2 but is not compared here with a newly trained architecture-matched supervised model on the same curriculum. Architecture and training-data effects are not isolated.
6. Authored simulator resource effects are not empirical FerrumOS runtime effects. The HUD boundary measurement observed zero normalized heap delta, so an authored non-zero simulator delta cannot be installed as runtime truth.
7. Brier and ECE values on the v3.4 final fixture describe deterministic scores, not calibrated probabilities. The balanced catalog and deterministic score levels limit their interpretation.
8. The original fixture found near parity between selected JEPA and a per-action mean baseline on binary decisions. Complete-pipeline variation is substantial.
9. Original runtime timing has 1 ms guest-timer resolution in one QEMU/WHPX environment. Outstanding previews are serialized, and provider latency, execution, confirmation, and physical I/O are excluded.
10. Boundary calibration covers one safe action class. No dangerous-recall or distribution-free calibration guarantee follows.
11. No independent human labels, completed natural-use study, adversarial operator study, formal verification, or independently designed external evaluation is reported.
12. v3.4 has no completed in-guest authority test, physical timing, actuator dynamics, sensor latency, or production rollout. The deployed model and manifest remain runtime-v2.
13. The original models reproduce only at the 24 BLAS threads of the development machine. Retrained at 4 or 1 thread, every model differs, the safe-replay false alarms and the rollout errors change, and the near-tie with the per-action mean reverses at 4 threads (Sections 6.2 and 8.2). The deployed digests are fixed, so measurements of the deployed files stand.

16.1 Valid claims

- FerrumOS implements a hybrid pre-execution mediation architecture in which learned output cannot erase deterministic policy or grant kernel capability.
- The public corpus, original evaluation artifacts, and release package are reproducible and digest-bound.
- The original selected JEPA predicts transitions better than its matched autoencoder, while its binary safety advantage over a per-action mean is not material on the authored fixture.
- The frozen v3.4 candidate improves runtime-v2 rollout error at all registered horizons on the untouched published test partition.
- Request-bounded v3.4 deterministic policy perfectly separates the new fixed deterministic catalog, with paired source-stratified uncertainty reported.
- No deployment or learned final-fixture safety advantage is claimed.

16.2 Invalid claims

- "JEPA makes FerrumOS safe."
- "The new model prevents real production incidents."
- "Zero observed final-fixture errors imply zero underlying error probability."
- "The v3.4 candidate is deployed or runtime-verified."
- "The incident-informed fixture is independent third-party validation."

17. Reproducibility and artifact checklist

Artifact or control	Status	Evidence
Published corpus ledger and episode split	Complete	13,697 rows; zero episode overlap
Original selected model pair	Complete	Versioned binaries and SHA-256 manifest
Five original complete pipelines	Complete	Seeds 17, 42, 91, 123, 2026 reported
Original safety fixture and predictions	Complete	500 cases; three-arm and baseline decisions
Runtime, concurrency, and failure JSON	Complete for runtime-v2	Ring-3 benchmark and disposable QEMU tests
Public dataset release	Complete	DOI 10.5281/zenodo.21829193; download verified
v3-v3.4 protocols and failures	Complete	Every selection/validation result retained
Final-catalog selection secrecy	Complete	Validation access audit records zero reads
v3.4 final evaluation	Complete	512 cases; source-stratified paired bootstrap
v3.4 deployed authority	External gap	Candidate archived, promotion disabled
Independent human labels and natural use	External gap	Registered workflow; no claimed outcome

17.1 Core reproduction commands

```
python scripts/verify_world_model_paper_evaluation.py
python scripts/evaluate_world_model_boundary_calibration.py
node scripts/benchmark_world_model_runtime.mjs --iterations 100
node scripts/verify_world_model_preview_concurrency.mjs
node scripts/evaluate_world_model_failure_modes.mjs
python scripts/verify_world_model_dataset_release.py target/world-model-dataset-release

python scripts/verify_world_model_incident_sources_v3.py --online
python scripts/select_world_model_jepa_v3_1.py
python scripts/select_world_model_jepa_v3_2.py
python scripts/validate_world_model_jepa_v3_3.py
python scripts/validate_world_model_jepa_v3_4.py
python scripts/evaluate_world_model_jepa_v3_4_final.py --dataset <public-jsonl>
python scripts/verify_world_model_jepa_v3_4.py --online
```

The final evaluator refuses to overwrite an existing final catalog. Selection and validation scripts install file-open audit hooks that deny final source and scenario paths. Candidate, protocol, selection, validation, scenario, and result artifacts are SHA-256 bound.

17.2 Verification state at v1.2 freeze

The runtime-v2 verifier passes 9/9 checks. The incident-source verifier passes 23/23 checks with official-source availability recorded. The v3.4 verifier passes 27/27 checks, reports `promotion_eligible: false`, and confirms deployed digests are unchanged.

18. Conclusion

FerrumOS demonstrates that an agentic operating system can place a provider-independent predictive screen at the action mediation boundary while preserving deterministic policy and kernel authority. The original study established the architecture, corpus, strong baselines, runtime cost, failure behavior, and limits of the first hybrid gate. Its strongest result was not JEPA superiority: a simple per-action mean was nearly tied on thresholded safety decisions, full training pipelines varied, and every false negative could be assigned to a concrete missing semantic or temporal mechanism.

The registered v3-v3.4 extension turns those failures into a narrower systems result. The frozen candidate improves untouched-corpus rollout prediction, while request-bounded deterministic authority removes the unsafe implication that hypothetical repeated rollout is equivalent to a currently authorized request. On the new source-held-out deterministic simulator, policy covers every registered case without safe-control intervention. Exact ablation shows that JEPA adds no safety decisions there.

The resulting contribution is therefore an evidence-disciplined authority architecture: learned forecasting can add caution and predictive detail, but exact policy, capabilities, and confirmation retain execution authority. Failed iterations, source partitions, digests, uncertainty, attribution, and non-promotion are part of the result. Deployment remains withheld until runtime authority, timing, and resource-effect gates pass.

Changes in version 1.3

Version 1.3 adds the dependence of the original trained models on the BLAS thread count, found by an exploratory audit after v1.2 (`docs/research/ferrumos_improvement_fi1_report.md`, detail in `docs/research/fi1_reports/pipeline_round3.md`, numbers in `docs/research/fi1_pipeline_round3.json`). No number of v1.2 is changed or recomputed, and no sentence or limitation is removed.

1. Section 6.2: the committed encoder and seed-17 transition model reproduce byte for byte only at 24 threads; the H=3 rollout error of the three transition models is 0.039 to 0.041 committed against 0.064 to 0.085 at 4 and 1 thread.
2. Section 8.2: the rules + JEPA confusion and balanced accuracy at 4 and 1 thread; the per-action-mean comparison (81.4% against 81.2%) becomes 80.8% against 81.2% at 4 threads; the comparisons with rules only, with the autoencoder, and the rules-against-combined interval hold at every count; safe-replay false alarms 216 of 1,969 against 50 and 58.
3. Limitation 13.

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